TSM 099 Quadratic and Rational Equations Test

Practice 2 | Version 202610122020.26

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right.

QUESTIONS	ANSWERS
1. Solve by factoring. $x^2 + 3x - 10 = 0$	1. Solutions 20 pts
2. Solve by taking square roots; do not expand the square. $2(x)^2 = 18$	2. Solutions 20 pts
3. Solve. List excluded values and check in the original equation. $\frac{6}{2x-4} + \frac{10}{x^2-7x+10} = \frac{4}{x-5}$	3. Excluded values 5 pts Solution/check 15 pts
4. Solve with the quadratic formula. Simplify radicals. $2x^2 + 2x - 3 = 0$	4. Solutions 20 pts
5. Solve. List excluded values and check candidates in the original equation. $\frac{6}{x^2-9} + \frac{x-4}{x-3} = \frac{7}{x+3}$	5. Excluded values 4 pts Candidates/check 6 pts Solution 10 pts

Instructor answer key: TSM 099 Quadratic and Rational Equations Test

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Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

TEST PAGE 1

1. Solutions

$x = -5,\ 2$

20 pts

2. Solutions

$x = -3,\ 3$

20 pts

3. Excluded values

$\{2, 5\}$

5 pts

Solution/check

$x = 3$

15 pts

4. Solutions

$x = -\frac{1}{2} + \frac{\sqrt{7}}{2},\ \ -\frac{\sqrt{7}}{2} - \frac{1}{2}$

20 pts

5. Excluded values

$\{-3, 3\}$

4 pts

Candidates/check

$x = 3$ (rejected), $x = 5$

6 pts

Solution

$x = 5$

10 pts